Is ‘empowerment’ just a fad? Control, decision-making, and information technology

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Invited paper

The logic in this paper shows why greater decentralization in business (including ‘empowerment’) is a response to fundamental changes in the economics of decision-making that are enabled by new information technologies. Our research suggests that a simple pattern of three successive stages underlies many of the changes that are taking place: As communication costs fall, independent decentralized decision-makers are replaced, first by centralized decision-makers, and then by connected decentralized decision-makers. This pattern explains important aspects of economic history in this century, and suggests that empowerment is not just a fad, but likely to become even more important in the next century.

The paper also suggests that our very notions of centralization and decentralization are incomplete. When most people talk about empowerment, they are only thinking about going ‘halfway’ toward what is possible. To fully exploit the possibilities of new information technologies, we may need to expand our thinking to include ‘radically decentralized organizations’ — like the Internet and market economies.

1. Introduction

Are you stifling innovation and creativity in your organization by trying to ‘micromanage’ too much? Or are you operating your organization like many autonomous fiefdoms and failing to get the benefits of really being ‘one company’? Should you be giving more autonomy to the people in your organization? Or do you need to take control and show ‘real leadership’?

As these questions illustrate, some of the oldest and most difficult problems in being a manager involve when and how to exercise control. And one of the central issues in creating the organizations of the next century will be finding the right way to balance top-down control and bottom-up empowerment [1]. For example, so much recent business rhetoric has focused on the importance of ‘empowering’ workers, that the term itself has already become an almost meaningless cliché. Is this talk of ‘empowerment’ just a fad, soon to pass the way of so many other business panaceas? Or is there something fundamental changing that makes more decentralized control in organizations increasingly desirable?

Our research suggests that the dramatically decreasing costs of information technology are, in fact, changing the economics of organizational decision-making to make decentralized control more desirable in many situations. Our work also suggests that our very notions of centralization and decentralization are incomplete. When most people talk about ‘decentralized organizations’ and ‘empowerment’, they are thinking about relatively timid shifts of power within what is still a fairly conventional hierarchical structure. These forms of ‘empowerment’, however, only go halfway toward what is possible. To fully exploit the possibilities of new information technologies, we may need to expand our thinking to include ‘radically decentralized organizations’ — like the Internet, markets of all kinds, and scientific communities — as models for new ways of organizing work in the 21st century.

Our research suggests that a simple pattern underlies many of the changes that will take place. As improvements in technology reduce the costs of communication and co-ordination, the most desirable way of making many decisions moves through three stages. In the first stage, when communication costs are high, the best way to make decisions is with independent decentralized decision-makers. Most economic decisions for most of human history have been made in this decentralized way by people in largely independent tribes, villages, and towns around the world.

But as the costs of communication fall, it becomes desirable in many cases to bring remote information together in one place where centralized decision-makers can

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see a more global perspective and make better decisions than isolated local decision-makers could. The economic history of the 20th century has been, in large part, a story of centralizing economic decision-making in large global corporations. And for many kinds of decisions, there are still substantial benefits of centralization yet to be exploited.

But as communication costs continue to fall even further, there comes a point for many decisions where connected decentralized decision makers can be even more effective than centralized ones. These decentralized, connected decision-makers can combine the best information available anywhere in the world with their own local knowledge, energy, and creativity. As our economy becomes increasingly based on creative innovation and ‘knowledge work’, and as new technologies make it possible to connect decentralized decision-makers on a scale never before possible in history, exploiting these opportunities for ‘empowerment’ may well be one of the most important themes in the economic history of the next century.

Of course, there are many factors other than communication costs that affect centralization and decentralization in organizations. Among the many factors that can be important, for instance, are patterns of interpersonal trust, locations of decision-relevant information, personal motivations, prior distributions of power within the organization, government regulations, national cultures, organizational traditions, and individual personalities [2].

In a given situation at a given point in time, various combinations of these other factors can be much more important than communication costs in determining where decisions are made. The goal of our work, however, was not to analyze the complex question of how all these factors interact in particular situations. We focused, instead, on a simpler question: ‘What is the logical relationship between reducing communication costs over time and the economics of different decision-making structures?’

Understanding this relationship is important for three reasons. First, it helps us understand conceptually what the economic effects of reduced communication costs would be if all other factors remained constant. Second, it provides a possible explanation for a variety of well-known facts, such as broad historical trends in organizational structures over the last century. Finally, to the degree you believe that reductions in communication costs enabled by IT are likely to be important in the future, our work suggests an effect they may have [3]. Whether this factor actually turns out to be important is, of course, uncertain. But if relentless improvements in information technology continue to reduce the costs of communication by a factor of 10 every few years in the future as they have in the past, our work suggests that shifts toward more decentralized empowerment will be a continuing fact of life in many organizations of the future.

2. How will information technology affect centralization and decentralization of organizations?

Ever since at least 1958, when Leavitt and Whisler [4] predicted that information technology would lead to the elimination of middle managers and to greater centralization of decision-making, people have speculated about how information technology will affect centralization and decentralization in organizations. And over the years, there have been numerous examples of changes in both directions. In some cases, information technology (IT) appears to have led to more centralization, in other cases to more decentralization, and in still others, it appears to have had no effect at all on centralization [5]. Based on previous research, therefore, there appears to be no clear answer to the question of how IT affects centralization and decentralization of organizations.

Much of the confusion in this discussion, however, results from lumping two kinds of decentralization together. When we distinguish between decentralized control by unconnected (i.e. independent) decision makers and decentralized control by connected decision makers [6], a clearer pattern begins to emerge. As described above, our research leads us to expect that unconnected decentralized decision-makers should be common when communication costs are very high. Then, as communication costs fall, centralized decision-making should become more desirable. Finally, as communication costs fall still further, connected decentralized decision-making should become desirable in many situations.

The basic logic of this progression derives from two simple assumptions:

- new information technologies will significantly reduce the costs of communication, and
- each stage in this progression requires more communication than the previous one, but — at least in some situations — has some other advantages over the previous stage.

Thus, as communication costs decrease, there will eventually come a point for each stage where its other advantages over the previous stage will be more important in some situations than its (diminishing) communication cost dizadvantage.

3. Explaining history

As mentioned above, this simple logic provides an explanation for some of the most salient aspects of economic history in this century. According to this interpretation, the dramatic rise of large organizations in the last hundred years or so was motivated, in part, by the economic benefits of centralized decision-making. For many kinds of decisions, centralized decision-makers can efficiently integrate diverse kinds of remote information and thus make better decisions than the unconnected (independent) local decision-makers they superseded [7]. And this centralized decision-making was itself rendered economically feasible by the advances in various information technologies (including not just computers, of course, but also telephones, televisions, radio, and so on). For much of this century, this was the 'main game in town', and, as recent megamergers illustrate, many managers believe there are still significant benefits from additional centralization in certain situations.

But in the latter part of this century, a different kind of change has also become increasingly important. Many companies are 'flattening' their organizations by removing layers of middle managers and increasing 'spans of control' of the remaining managers. In many of these cases, managers who are now supervising significantly more people than their predecessors are forced to delegate more important decisions to their subordinates. In fact, 'empowerment' of employees has become one of the most popular buzzwords of the 1990s. More and more employees are finding themselves with increasing responsibilities, and more and more managers are trying to act like 'coaches' who help their employees solve problems, rather than decision-makers who give orders and monitor compliance.

In addition, more work is being co-ordinated outside the boundaries of traditional hierarchical organizations altogether. Large companies are outsourcing more and more of their non-core activities. In many industries, small companies are playing an increasingly important role. And 'virtual corporations', 'networked organizations', and other shifting alliances of people and organizations are performing more and more of the work that might previously have been done inside a single large organization.

Our simple logic provides an explanation for these trends as well. Making decisions closer to the point where they are actually carried out (e.g. 'closer to the customer') can have many advantages. For instance, in many kinds of work, people are likely to be more energetic and creative if they have more autonomy about how they work and what they do. And 'local' decision-makers may also have access to information that is critical to making good decisions (e.g. the customers' unstated preferences) but hard to communicate to central decision-makers. These, then, are economic motivations for decentralizing decision-making.

But, in order for decentralized decision-makers to make intelligent decisions, they may also need to take into account the diverse kinds of remote information that enabled centralized decision-makers to make better decisions in the first place. And it is precisely the communication of this large amount of information to much larger groups of decision-makers that information technology now makes possible at a cost and on a scale never before seen in history.

Our logic suggests, therefore, that the recent trends toward decentralization in many organizations are not just a fad, but are instead part of an underlying dynamic in the economics of organization that is likely to become even more important in the next century.

Notes

1 See, for example: Johansen B, Saveri A and Schmid G: '21st century organizations: reconciling control and empowerment', Menlo Park, California, Institute for the Future (1995).

2 See, for example:

DiMaggio P J and Powell W W: 'The iron cage revisited: Institutional isomorphism and collective rationality in organizational field', American Sociological Review, 48, pp 147—160 (1983).

Galbraith J R: 'Organization design', Reading, Massachusetts, Addison-Wesley (1977).

Gurbaxani V and Whang S: 'The Impact of information systems on organizations and markets', Communications of the ACM, 34, No 1, pp 59—73 (1991).

Huber G P and McDaniel R R: 'The decision-making paradigm of organizational design', Management Science, 32, No 5, pp 572—589 (1986).

Markus M L: 'Power, politics, and MIS implementation', Communications of the ACM, 26, pp 430—444 (1983).

Schein E H: 'Organizational culture and leadership', San Francisco, Jossey-Bass (1985).

Scott W R: 'Organizations: rational, natural, and open systems', Third Edition, Englewood Cliffs, Prentice-Hall (1992).

Thompson J D: 'Organizations in Action', New York, McGraw-Hill (1967).

3 Our model is particularly intriguing in this regard because, unlike previous models of this question (e.g. Gurbaxani V and Whang S: 'The impact of information systems on organizations and markets', Communications of the ACM, 34, No 1, pp 59—73 (January 1991)), it shows how a simple model can explain changes in both directions while nevertheless predicting a broad long-run change in one direction.

4 Leavitt H J and Whisler T L: ‘Management in the 1980’s’, Harvard Business Review, 36, pp 41—48 (November/December 1958).

5 For summaries of previous research, see, for example: Attewell P and Rule J: ‘Computing and organizations: What we know and what we don’t know’, Communications of the ACM, 17, No 12, pp 1184—1192 (December 1984), and George F J and King J L: ‘Examining the computing and centralization debate’, Communications of the ACM, 34, No 7, pp 63—72 (July 1991).

6 For a previous paper making this distinction, see: Anand K S and Mendelson H: ‘Information and organization for horizontal multimarket co-ordination’, Stanford University, Graduate School of Business Research Paper No 1359 (October 1995).

7 See, for example: Chandler A D: ‘The visible hand: the managerial revolution in American business’, Cambridge, MA, Belknap Press (1977).

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